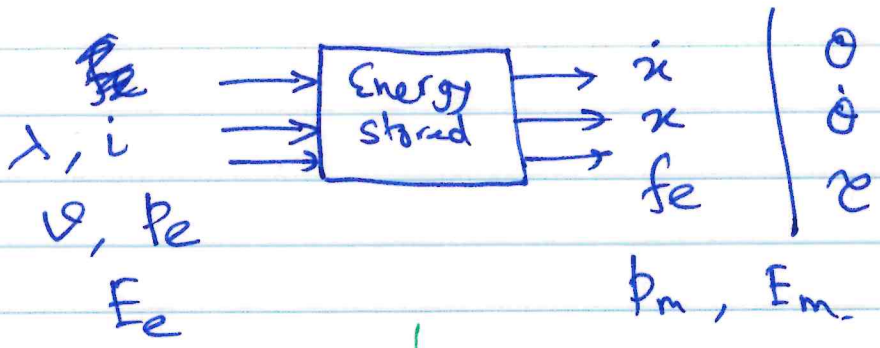
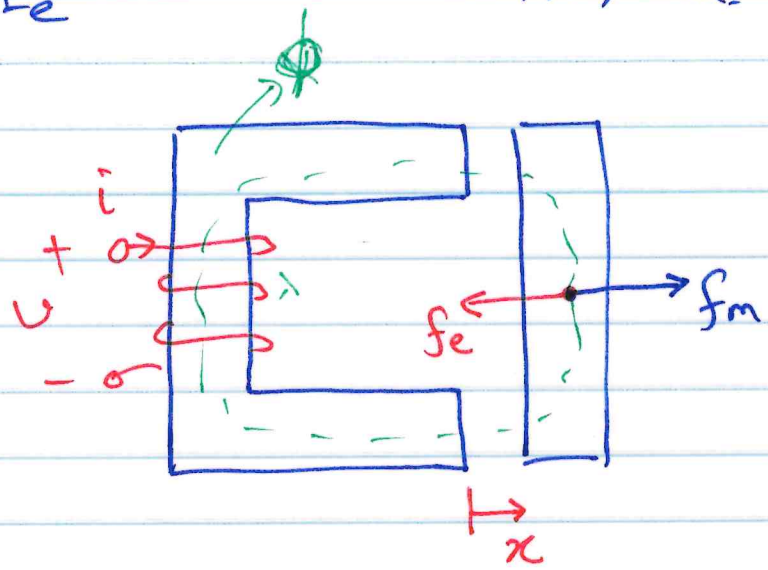
Lecture X





$$dW_m = i\,d\lambda - f_e\,dx$$

$$dW_m(\lambda, x) = \frac{\partial W_m}{\partial \lambda}\cdot d\lambda + \frac{\partial W_m}{\partial x}dx$$

$$\frac{\partial W_m}{\partial \lambda} = i \longrightarrow x\text{-const}$$

$$\frac{\partial W_m}{\partial x} = -f_e \longrightarrow \lambda\text{-const}.$$

Calculate f_e :

1. Magnetic Eq. Ckt.
2. $R_{eq}(x)$
3. $L(x)$

$$\boxed{\lambda = L(x) \cdot i}$$

$$W_m(\lambda, x) = \int_0^{\lambda} i' \, d\lambda' \quad \Big| \; x \text{ const.}$$

$$= \int_0^{\lambda} \frac{\lambda'}{L(x)} \cdot d\lambda'$$

$$\boxed{W_m(\lambda, x) = \frac{1}{L(x)} \cdot \frac{\lambda^2}{2}}$$

$$W_m = \frac{1}{L(x)} \cdot \frac{(L(x)\, i)^2}{2}$$

$$= \frac{1}{2} L(x)\, i^2$$

$$\downarrow$$

$$\boxed{\frac{\partial W_m}{\partial x} = \frac{1}{2} i^2 \frac{\partial}{\partial x}\left(L(x)\right)}$$

$$f_e(\lambda, x) = \frac{\partial}{\partial x}\left[\frac{1}{L(x)} \cdot \frac{\lambda^2}{2}\right] \quad \Big| \; \lambda \text{ const}$$

$$= \frac{\lambda^2}{2} \cdot \frac{\partial}{\partial x}\left[\frac{1}{L(x)}\right]$$

$$= \frac{-1}{2} L(x)^2 i^2 \frac{\partial}{\partial x}\left(\frac{1}{L(x)}\right)$$

λ, i $\qquad$ (x, fe)

$\lambda, x \rightarrow$ independent variable

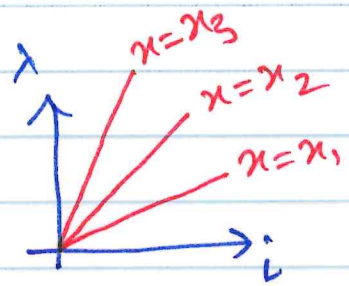
$\downarrow$

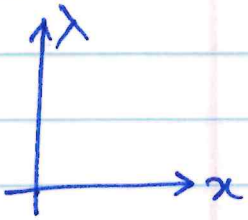
ω_m

$i, x \rightarrow$ independent variable.

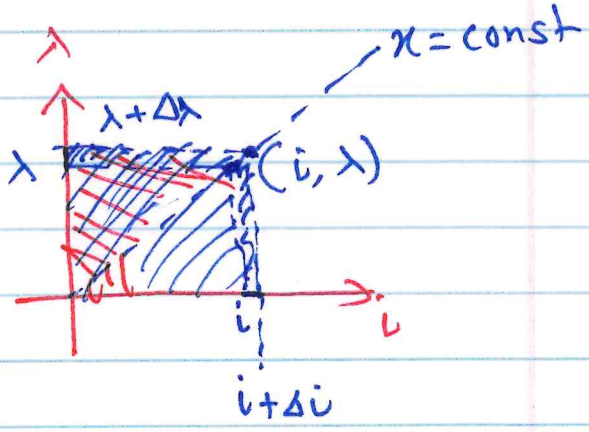
$\downarrow$

?





$\lambda = L(x) \cdot i$



$$(\lambda+\Delta\lambda)(i+\Delta i) = \lambda i + i\Delta\lambda + \lambda\Delta i + \Delta\lambda\Delta i$$

$$+ \Delta\lambda\Delta i \to 0$$

$$d(\lambda i) = i\,d\lambda + \lambda\,di$$

$$\boxed{(\lambda+\Delta\lambda)(i+\Delta i) - \lambda i} = i\Delta\lambda + \lambda\Delta i$$

$$dW_m(\lambda, x) = i\,d\lambda - f_e\,dx$$

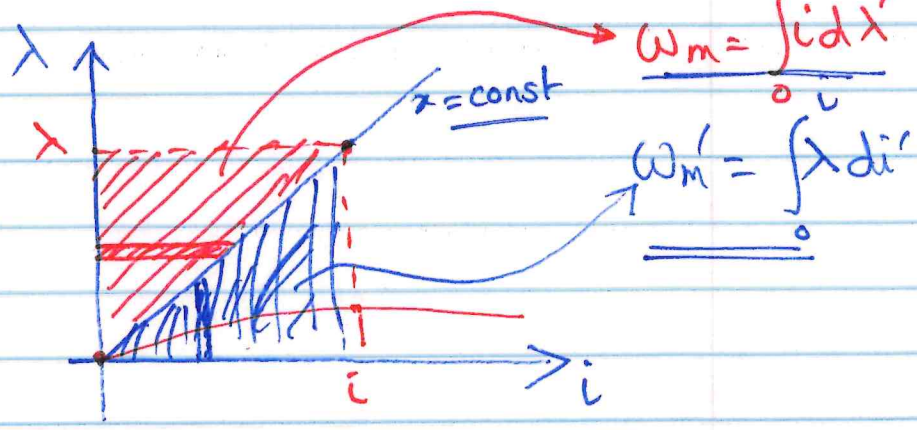
~~$dW_m(i, x) =$~~

~~$W_m(\lambda, x) = \int i\,d\lambda = \int f_e\,dx$~~

$$dW_m(\lambda, x) = \left[d(\lambda i) - \lambda\,di \right] - f_e\,dx$$

$$\lambda\,di + f_e\,dx = d\underbrace{\left(\lambda i - W_m(\lambda, x) \right)}_{W_m'(i, x)}$$

$$dW_m'(i, x) = \lambda\,di + f_e\,dx$$



$$\Rightarrow d W_m'(i,x) = \lambda\, di + f_e\, dx$$

$$\Rightarrow d W_m'(i,x) = \left.\frac{\partial W_m'}{\partial i}\right|_{x=\text{const}} \cdot di + \left.\frac{\partial W_m'}{\partial x}\right|_{i=\text{const}} \cdot dx$$

$$\left.\frac{\partial W_m'}{\partial i}\right|_{x=\text{const}} = \lambda$$

$$\boxed{\left.\frac{\partial W_m'}{\partial x}\right|_{i=\text{const}} = f_e}$$

$W_m'(i,x) \rightarrow$ coenergy

$W_m(\lambda,x) \rightarrow$ energy

$\lambda i = W_m(\lambda,x) + W_m'(i,x)$.

$$\boxed{\lambda = L(x)\, i}$$

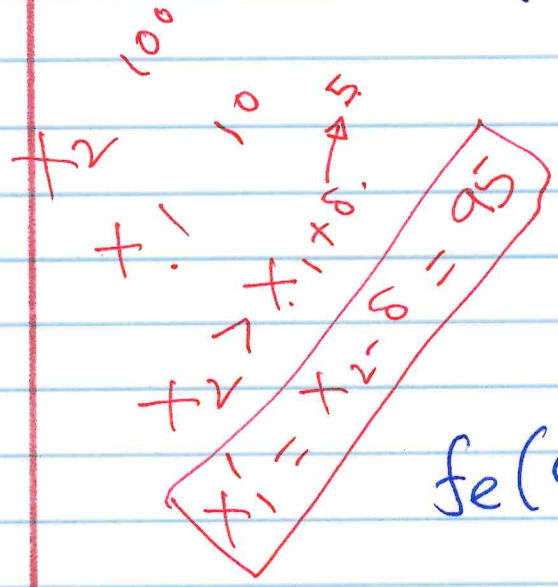
$$W_m'(i,x) = \int_0^i \lambda \, di'$$

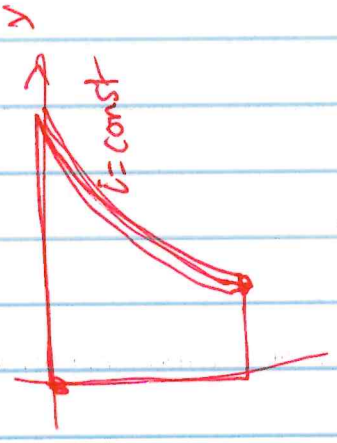
$$= \int_0^i L(x) \cdot i' \, di'$$

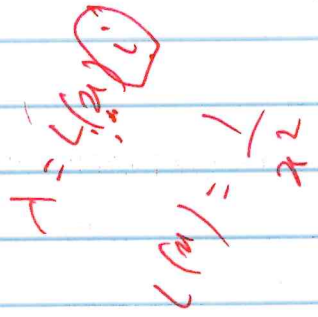
$$= \frac{1}{2} L(x)\, i^2$$

$$f_e(i,x) = \frac{\partial}{\partial x}\left[\frac{1}{2} L(x)\, i^2\right]\Bigg|_{i=\text{const}}$$

$$= \frac{i^2}{2} \frac{\partial [L(x)]}{\partial x}$$





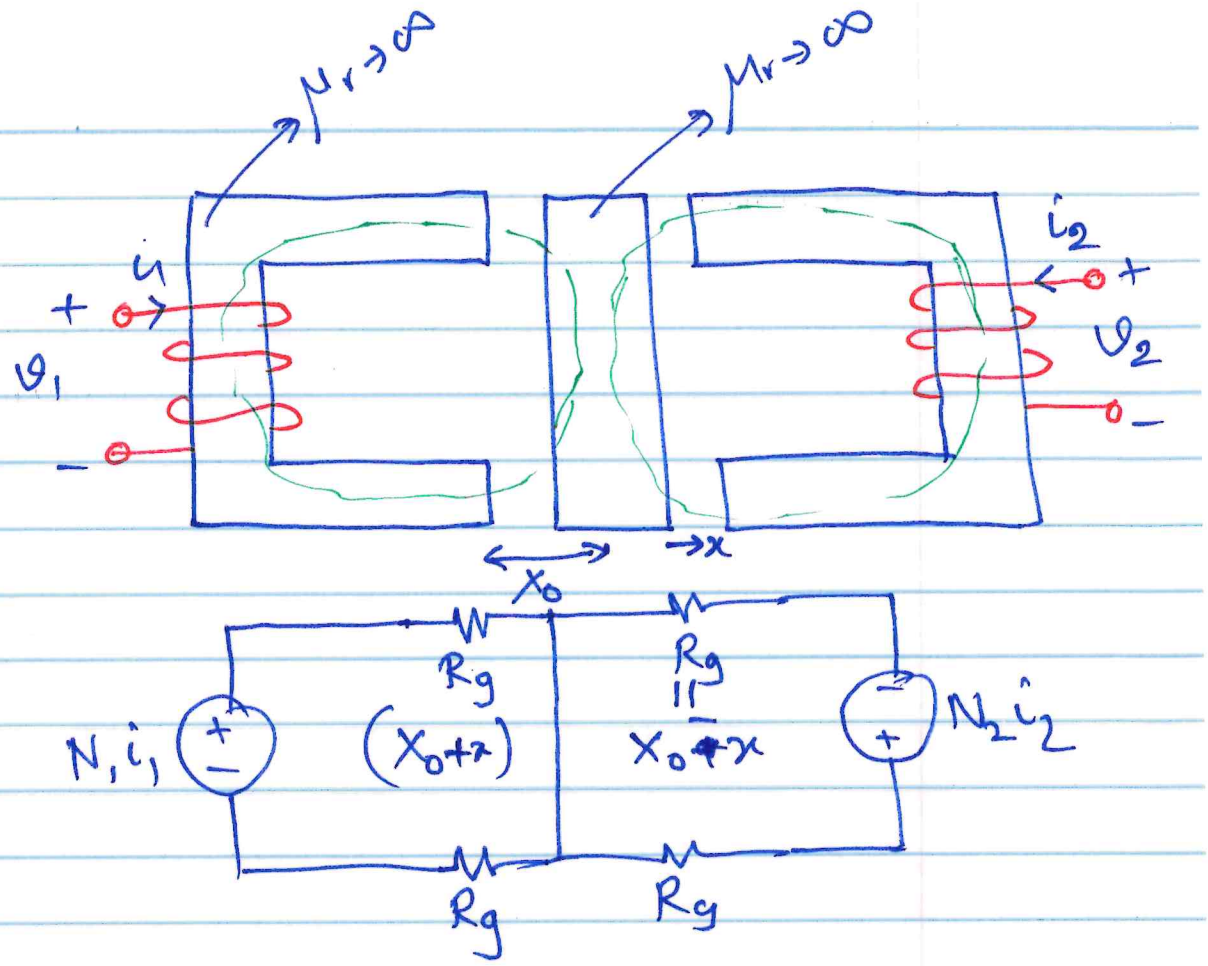


$$f_e = -\left.\frac{\partial W_m(x,\lambda)}{\partial x}\right|_{\lambda = \text{const}}$$

$$f_e = +\left.\frac{\partial W_m'(x,i)}{\partial x}\right|_{i = \text{const}}.$$

$$f_e = -\frac{\lambda^2}{2}\frac{\partial}{\partial x}\left[\frac{1}{L(x)}\right]$$

$$f_e = \frac{i^2}{2}\frac{\partial}{\partial x}\left[L(x)\right]$$


$\mu_r \to \infty$
$\mu_r \to \infty$
i_1
+
v_1
−
i_2
+
v_2
−
x_0
$\to x$
R_g
R_g
$N_1 i_1$
(x_0+x)
x_0+x
$N_2 i_2$
R_g
R_g